

Efficient glycolysis of recycling poly(ethylene terephthalate) via combination of organocatalyst and metal salt

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ABSTRACTS

The recycling of discarded polyethylene terephthalate (PET) is an effective way of catering to global sustainable economic development strategies. It is well known that catalysts play a crucial role in the degradation of PET, based on the fact that most of the currently available catalysts have some disadvantages, including complex preparation process, severe reaction conditions and insufficient reaction efficiency. Herein, we report a combination of metal catalyst and organoazacyclic 1,8-diazabicyclo [5.4.0]undec-7-ene (DBU) as cooperating catalysts were used to depolymerize waste polyethylene terephthalate into bis(2-hydroxyethyl) terephthalate (BHET) directly. Screening of several metal salts under optimal conditions showed that zinc acetate ($\text{Zn}(\text{OAc})_2$) was the most effective co-catalyst. The waste PET could be completely depolymerized at 180 °C for 77 min with the weight ratio of EG/PET was 3:1 and the molar ratio of $\text{Zn}(\text{OAc})_2$ /DBU was 1:2, and the content of BHET in the recrystallized product reached 78.2%. Kinetic studies showed that the activation energy of depolymerization of PET was 115.48 kJ/mol. It was speculated that the hydrogen bond formed by $\text{Zn}(\text{OAc})_2$ /DBU and EG exhibited excellent catalytic effect in the glycolysis process. Therefore, it is hopeful that this simple and mild composite catalyst can provide industrial ideas for recycling other waste polyesters.

1. Introduction

The annual production of plastics and man-made fiber is increasing at an astonishing rate because of its convenience and excellent performance. Among them, polyethylene terephthalate (PET), a semi-crystalline polymer, which is favored for its excellent processing properties and its own good resistance in packing materials and textiles [1,2]. The annual production of PET has exceeded 33 million tons and will continue to grow [3]. However, the wide range of applications will inevitably bring different levels of pollution, according to incomplete statistics, the global discharge of waste polyesters reached 79.92 million tons in 2020. More than 80% of these waste resources are buried in garbage landfill or the natural environment [2,4], due to its inert molecular structure, it takes hundreds of years to degrade into small molecules completely or other nontoxic substances. The accumulation of large amount of waste polyesters has aggravated resources depletion and environmental pressure. Therefore, how to effectively recycle and reuse waste PET polyester has become the focus of attention for researchers at present.

The recycling disposal of PET has undergone iterative development

of physical methods [5], machining methods [6], chemical recycling [7–10] and then biotechnology enzymatic degradation [11,12], glycolysis of PET is the most frequent method to recover waste polyesters. It uses the reversibility of the chemical reaction to convert it into the monomer bis(2-hydroxyethyl) terephthalate (BHET), which was repolymerized into PET or other polyesters [13]. The most widely catalyst for chemical glycolysis of PET was zinc acetate ($\text{Zn}(\text{OAc})_2$), which was reported to be an effective catalyst for molecules containing an ester structure [8]. Vaidya et al. [14] used $\text{Zn}(\text{OAc})_2$ for the first time as a catalyst for glycol depolymerization of PET, and the obtained product of BHET was purified and repolymerized to new polyesters. Ghaemy et al. and Xi et al. performed a comparative analysis of metal acetate-based catalysts in glycolysis PET using $\text{Zn}(\text{OAc})_2$, $\text{Mn}(\text{CH}_3\text{COO})_2$, $\text{Co}(\text{CH}_3\text{COO})_2$ and $(\text{CH}_3\text{COO})_2\text{Pb}$ as catalysts, respectively, with the measured reaction activity of catalysts was ranked as $\text{Zn}^{2+} > \text{Mn}^{2+} > \text{Co}^{2+} > \text{Pb}^{2+}$. The depolymerization reaction reached equilibrium after 2–3 h of reaction at 196 °C with the yield of BHET was 85.6% [15,16]. The alkali metal salts [17,18] such as sodium carbonate, sodium bicarbonate, sodium sulfate and potassium sulfate as catalysts for PET glycolysis, the yield of BHET was 62.18% after reacting at 190 °C for 8 h.

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The obvious disadvantages in these research are that the long reaction time and the high catalyst content can lead to the impurity of the subsequent product.

As the degradation of waste polyesters has been intensively studied, new catalysts have been applied to the degradation of PET, such as ionic liquids [19–21], deep eutectic solvents [22], superparamagnetic nanoparticles [23] and nanocomposite [24]. Different from metal catalysts, these are need to go through a complex preparation process before glycolysis, and the heavy metal-based catalysts can cause impurity of product and environmental pollution.

Organocatalysts have been developed as a green alternative to conventional metal catalysts, Fukushima et al. explored cyclic amidine catalyst for glycolysis of PET into BHET monomers, which has an unexpected efficiency in glycolysis reactions [25]. Urea are used as catalysts or deep eutectic solvents in PET glycolysis [26]. However, they embody similar drawbacks with low yield of the product BHET and hindering industrial applications.

It was found that several different types of catalysts have their advantages and drawbacks for the glycolysis of polyester, such as high reaction temperature and long reaction time, even the residual amount of metal catalyst is high, which is unfavorable for the subsequent industrial development. The current recycling technology with ultrafast reaction rate under mild condition is still a significant and challenging work. Therefore, other methods were sought to commercialize PET glycolysis. This work was inspired by the degradation of PET under mild conditions by a composite catalyst of basic nitrogen heterocycle and metal salt capable of acting as Lewis acid [27]. Organocatalysts are expected to be popular catalysts to solve some drawbacks [28], whether the effective combination of two different types of catalysts can play a significant role in promoting the degradation of polyester. Herein, the metal catalyst zinc acetate and the organic compound 1,8-diazabicyclo [5.4.0]undec-7-ene (DBU) were selected to degrade waste polyester, both the effect of composite catalyst on PET glycolysis and the effect of different reaction parameters on the glycolysis reaction of PET glycolysis was investigated, with the glycolysis products were characterized and analyzed, expecting that this composite catalysts can be useful for the industrial recovery of polyesters.

2. Experimental section

2.1. Materials

Polyethylene terephthalate (PET) was obtained from 500 ml post-consumer bottle of Nongfu Spring Co., Ltd., Ethylene glycol (EG, AR, 99%), zinc acetate (AR, 99%), magnesium acetate (AR, 99%), 1,8-diazabicyclo [5.4.0]undec-7-ene (DBU, AR, 99%), sodium ethoxide (AR, 96%), all from Shanghai Aladdin Biochemical Technology Co., Ltd.; bis (2-hydroxyethyl terephthalate) (BHET, > 85%), titanium phosphate ($\text{Ti}_3(\text{PO}_4)_4$), potassium carbonate (AR, 99%) from Shanghai Macklin Biochemical Technology Co., Ltd. All reagents and materials were purchased commercially and used without further purification.

2.2. Pretreatment of PET before glycolysis

Pretreatment of discarded mineral water bottles: The discarded mineral water bottles were collected, cutting into small pieces of 5×5 mm after removing the cap, bottom and label paper, cleaned with alcohol and sonicated for 30 min to remove impurities and contaminants from the surface of waste bottle, then placed in an oven at 60 °C for 12 h to dry, ready for use.

2.3. Procedure for the catalytic degradation of PET

In the typical PET glycolysis reaction, 5.0 g PET fragments, 15 g EG and an appropriate amount of zinc acetate was added into a four-necked flask equipped with a condenser tube, a thermometer, started the me-

chanical stirrer slowly until the temperature heated to the specified temperature, then added 0.2 mmol of 1,8-diazabicyclo [5.4.0]undec-7-ene subjected to glycolysis. The PET flakes were fully degraded after being reacted at 180 °C for 77 min. After the reaction was completed, a large amount of deionized water was added to the filtrate when the reaction solution cooling down to 80 °C. The solution and filtered quickly to separate the unreacted PET(w_1) and water-insoluble oligomers. And collected filtrate and put it in a refrigerator at 4 °C for 24 h to crystallize, suction filtration and drying to obtain the white needle-like product of BHET. The calculation formulas for the conversion of waste PET and the yield of product BHET are as follows:

$$\text{Conversion of PET} = \frac{w_0 - w_1}{w_0} \times 100\% \quad (1)$$

$$\text{Yield of BHET} = \frac{w_{\text{BHET}}/M_{\text{BHET}}}{w_0/M_{\text{PET}}} \times 100\% \quad (2)$$

Where w_0 is the initial weight of PET and w_1 is the weight of the unreacted PET. w_{BHET} is the weight of BHET, M_{BHET} is the molecular mass of BHET (254 g/mol), M_{PET} is the molecular mass of PET repeatunit (192 g/mol).

2.4. Characterization

The advance-400 MHz nuclear magnetic resonance spectrometer of bruker was used to test the NMR hydrogen spectra of different recrystallization products, with dimethyl sulfoxide (DMSO- d_6) as the solvent, Nicolet 5700 Fourier Transform Infrared Spectrometer was used to test the alcohol solution functional groups at different wavelengths, and the KBr tableting method was used to test the spectral scanning range of 500–4000 cm^{-1} with the resolution was 4 cm^{-1} .

The thermogravimetric test was carried out by the Mettler Toledo TGA, Swiss, samples of approximately 5–8 mg and tested under a nitrogen atmosphere, the nitrogen gas flow rate was 45 ml/min, and the heating rate of 10 °C/min was from 25 °C to 600 °C. The DSC test of the product was measured by Mettler Toledo DSC 3 in Switzerland to weigh 5–10 mg of samples, heating samples from room temperature to 300 °C at a rate of 10 °C/min under a nitrogen atmosphere. The surface morphology of the product was analyzed by the Gemini 500 Scanning Electron Microscope of the Carl-Zeiss company in Germany, and the electron microscope accelerated voltage to 30 kV, the picture was enlarged with the inlens secondary electron detector in the lens barrel. The ultra-efficient polymer chromatography (APC) test used acquity APC columns, tetrahydrofuran (THF) as the mobile phase, the flow rate of 0.5 ml/min, and column oven and indication detector temperature was 45 °C, the concentration of the sample was 4 mg/ml and PMMA was selected as the standard sample for calibration. The ultra performance liquid chromatography mass spectrometry (UPLC-MS), column temperature was 30 °C, sample chamber temperature was 25 °C, mobile phases were water (phase A) and acetonitrile (phase B), mobile phase flow rate was 0.3 mL/min.

3. Results and discussion

3.1. Screening of metal salts for glycolysis

Metal catalysts have unique advantages and different metal cations have different ways of destroying PET. Fig. 1 showed the product distributions for the catalytic glycolysis of PET at 180 °C for 77 min in the presence of various metal salts, according to the literature, it took 1 h for the glycolysis of PET at 196 °C to gradually break down the macromolecules into small molecules of BHET [16,29]. In this paper, metal salt catalysts and DBU is combined for glycolysis of PET for the first time, the conversion rate of PET and the yield of product BHET were compared and analyzed. It was found that the compound catalytic effect of zinc acetate and DBU has the best catalytic effect, and the purity of the

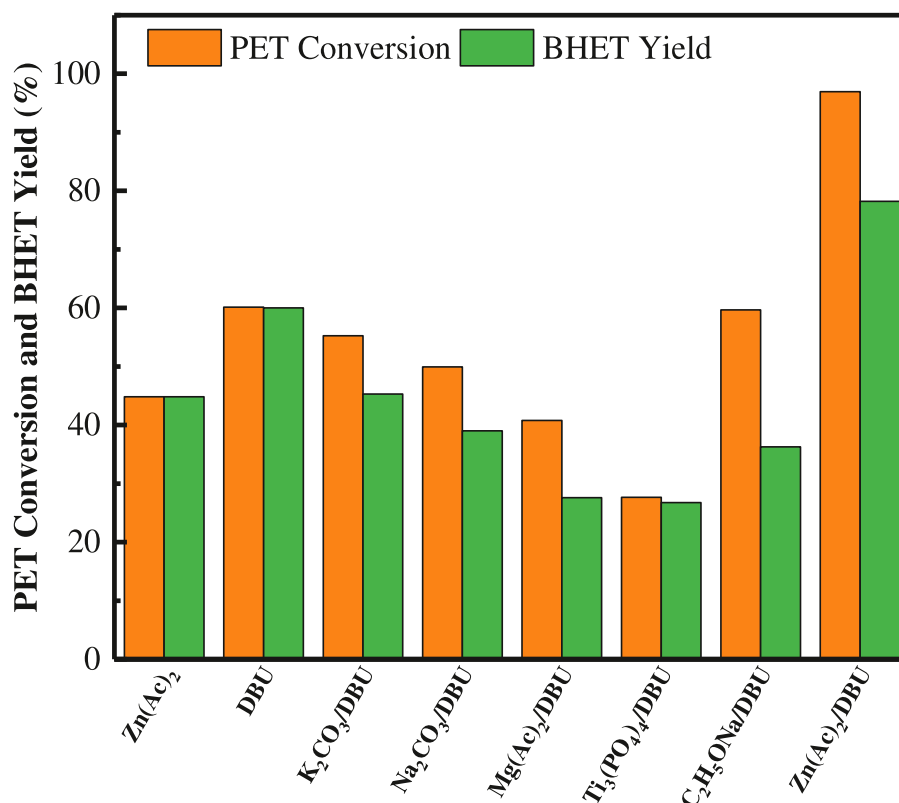


Fig. 1. Different catalysts of glycolysis of PET at 180 °C for 77 min, composite catalyst were Zn(OAc)₂, DBU, K₂CO₃/DBU, Na₂CO₃/DBU, Mg(Ac)₂/DBU, Ti₃(PO₄)₄/DBU, C₂H₅ONa/DBU, Zn(OAc)₂/DBU, Reaction conditions: PET (5.0 g), EG (15 g), catalyst (n(metal salts)/n(DBU) = 1:2).

product after recrystallization was as high as 94.85%.

Under the same conditions, 60% of BHET was obtained when using DBU alone for PET degradation. Similarly, 45% of BHET was acquired with zinc acetate catalyst. It has been reported that over 75% of BHET can be obtained by glycolysis of PET using 1 wt% zinc acetate at 198 °C for 8 h [15]. According to the public work, PET was degraded in 22 min at 190 °C with 10 mol% DBU for 0.5 wt% loading relative to PET [25].

Compared to other metal catalysts compounded with DBU, we speculate that the excellent catalytic effect of the zinc acetate composite catalyst is attributed to the possibility that the acetate and DBU can be seen as hydrogen bond acceptors of ethylene glycol activating for the

ulterior nucleophilic attack to a PET ester bond, while zinc is Lewis acid that can interact with carbonyl groups, which accelerates the breaking of the molecule chain [30–32].

3.2. Structural characterization of recrystallized products

Fig. 2(a) displayed the ¹H-NMR spectra of the recrystallization product. The molecular structure of BHET has four hydrogen atoms of different environments, of which 8.14 ppm was the resonance peak of the hydrogen atom on the benzene ring, and the hydrogen atom of the hydroxyl group (-OH). The atomic peak appeared at 4.91 ppm, 4.34 ppm

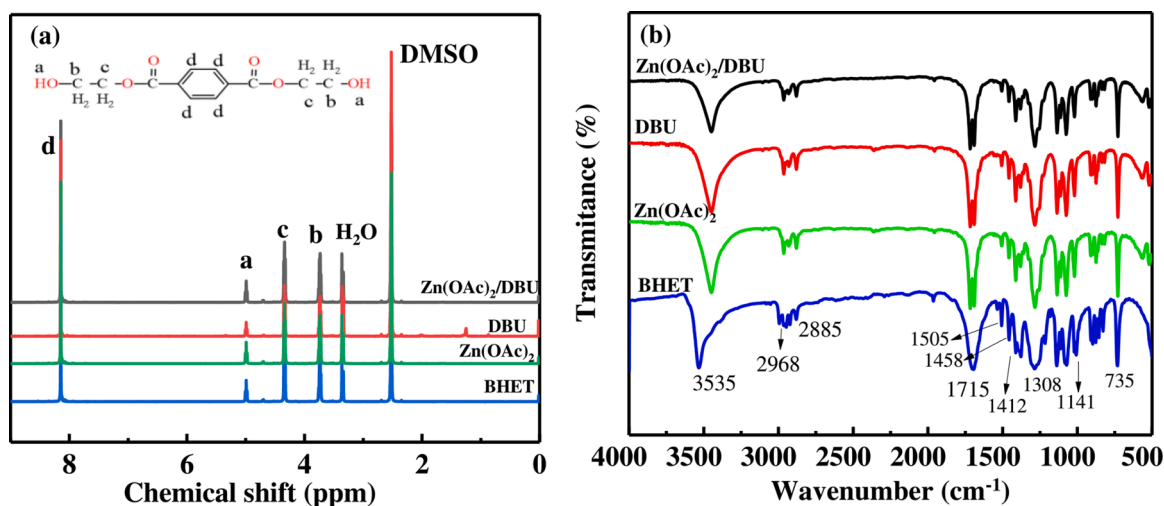


Fig. 2. (a), ¹H-NMR spectra of the recrystallization product. (b), Infrared spectra of the recrystallization product with Zn(OAc)₂, DBU, Zn(OAc)₂/DBU catalysts and commercial BHET.

was the peak of hydrogen atoms attached to the methylene ($-\text{CH}_2-$) and carbonyl ($-\text{C}=\text{O}$) groups, the methylene ($-\text{CH}_2-$) and hydroxyl ($-\text{OH}$) groups occurred at 3.97 ppm, which verified the structure of the recrystallized product.

The Infrared Spectra of the recrystallization products were shown in Fig. 3(b), the O-H stretching vibration in the hydroxyl group has absorption peaks in the region of $3700\sim 3200\text{ cm}^{-1}$, 2968 cm^{-1} , and 2885 cm^{-1} , the absorption peaks at 2885 cm^{-1} and 1458 cm^{-1} corresponded to the stretching vibration and bending vibration of methylene C-H, respectively, which indicated the existence of methylene- CH_2- . And the strong absorption at 1715 cm^{-1} and 1141 cm^{-1} were the stretching vibrations of $\text{C}=\text{O}$ and $\text{C}-\text{O}$, the absorption bands at 1505 cm^{-1} , 1458 cm^{-1} and 1308 cm^{-1} were accompanied with the stretching vibrations of $\text{C}=\text{C}$ and the in-plane bending vibrations of the benzene ring at $900\sim 675\text{ cm}^{-1}$. The absorption band at 735 cm^{-1} was caused by the C-H out-of-plane bending vibration on the benzene ring. The IR spectra of the recrystallization products of the $\text{Zn}(\text{OAc})_2/\text{DBU}$ composite catalyst was not significantly different from those of the single-component degradation.

To verify the content of other oligomers in the recrystallization products, it was displayed in the Fig. 3(a), which showed that different components of the recrystallization product with APC chromatogram. It could be seen from the table that the content of low polymer in the recrystallized product obtained by $\text{Zn}(\text{OAc})_2/\text{DBU}$ composite catalyst was lower than that the product obtained by single component degradation of PET. This was ascribed to the DBU and $\text{Zn}(\text{OAc})_2$ jointly accelerated the destruction of ester bonds on the molecules of PET, while -OH of ethylene glycol attacks the broken ($-\text{COO}$), which made the PET molecular chain broke more thoroughly. The content of BHET in the recrystallized product was as high as 94.85% are listed in Table 1, with only a small amount of dimer. Oligomers of PET degradation was deeply investigated by Fukushima et al. [33]. Analysis in combination with UPLC-MS mass spectrogram in Fig. 3(b). the ions (m/z) detected with TOF-MS- EI^+ were determined to be pseudo-molecular ion $[\text{M}+\text{Na}]^+$, molecular ion of BHET showed up as $m/z = 277.0682$ and was consistent with the molecular weight of BHET after being ionized by Na^+ . This result was matched with the retention time and molecular ion of standard materials. However, there are many oligomers in the recrystallized products from the degradation of other single components, which further indicates that the purity of the purified products from the two-component catalyst is high.

Table 1

Component analysis of recrystallized products obtained by degradation of different glycolysis catalysts.

Catalysts	BHET (%)	Dimer (%)	Trimer (%)	Tetramer (%)	Oligomer (%)
Commercial BHET	94.37	4.03	1.6	0	0
$\text{Zn}(\text{OAc})_2$	93.55	5.79	0.66	0	0
DBU	37.91	49.78	8.83	2.54	0.94
$\text{Zn}(\text{OAc})_2/\text{DBU}$	94.85	5.15	0	0	0

3.3. Thermal analysis of recrystallization product

In order to demonstrate the melting point and thermal stability of the recrystallization products, differential scanning calorimetry (DSC) and thermogravimetric analysis (TG) were performed. The recrystallization products under different catalysts can be revealed in Fig. 4(a), the product has a sharp absorption peak at 110°C , which can be inferred to be the melting point of the product [34,35]. Fig. 4(b) and (c) were the TG and DTG curves of BHET and glycolysis products, it could be seen from Fig. 4(b) that there were two weight loss platforms, this phenomenon was attributed to the thermal decomposition reaction when the weight loss stage was around $230\sim 290^\circ\text{C}$, which removed small molecules and decomposition of oligomers. The second weight loss occurred at around 436°C with a mass loss rate of 40%, which was caused by the thermal degradation of BHET polymerization into PET during the polycondensation process [36,37]. The different loss rates of the different products and BHET at the first weight loss were due to the different content of oligomers contained in the recrystallized product, resulting in different levels of product loss during the process of heating [20].

3.4. Analysis of cross-sectional morphology of samples during glycolysis

To investigate the glycolysis process of PET in EG, SEM morphology analysis was performed on the remaining PET under different reaction times. The SEM and structural characteristics of PET after 20 min, 40 min, 60 min and 77 min in Fig. 5, the PET surface was presented a rod-shaped regular arrangement. With the extension of the reaction time, the fracture cracks of the PET surface increased and gradually degraded into oligomers and BHET. It could be noticed visually that PET was gradually fractured from the surface and the fractured molecular polymers was rapidly depolymerized and eventually converted into BHET monomer.

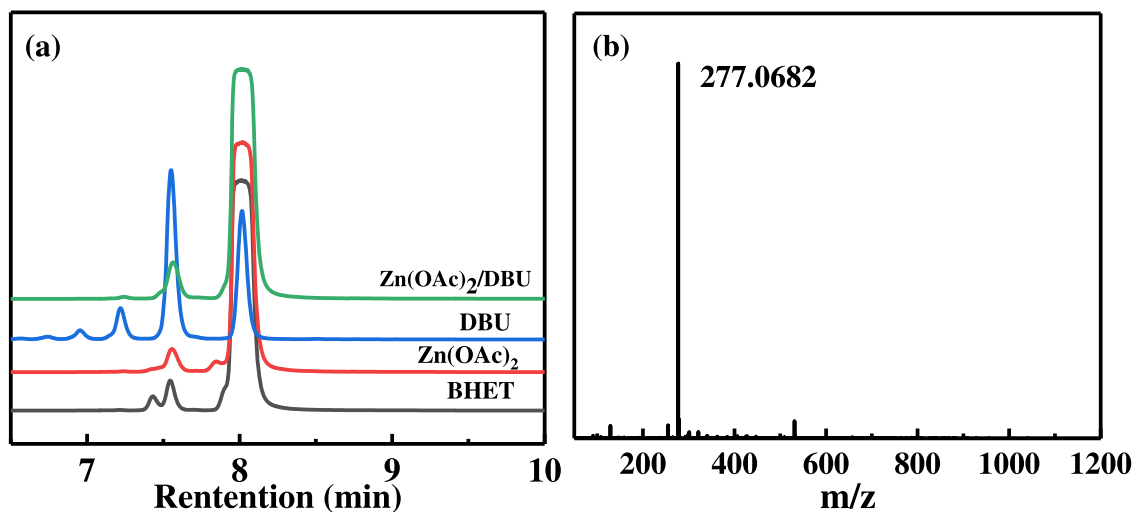


Fig. 3. (a), APC chromatogram of the recrystallization product under different catalysts with $\text{Zn}(\text{OAc})_2$, DBU, $\text{Zn}(\text{OAc})_2/\text{DBU}$ catalysts and commercial BHET. (b), HPLC-MS chromatogram of the recrystallization product with the $\text{Zn}(\text{OAc})_2/\text{DBU}$ catalyst.

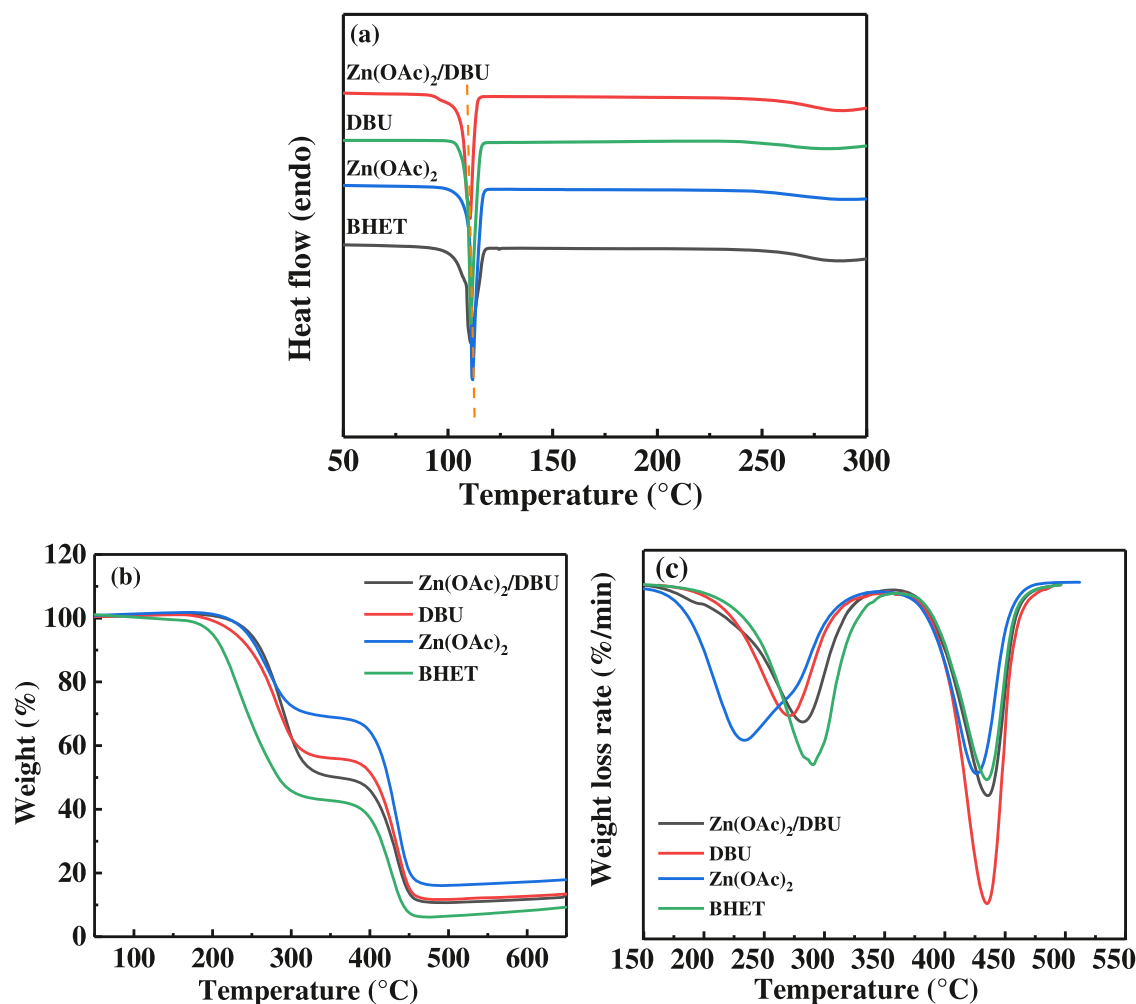


Fig. 4. thermal analysis curves of the recrystallization product: (a), DSC. (b), TG. (c), DTG.

3.5. Optimization of depolymerisation conditions

In order to explore the optimization conditions of PET glycolysis in the present catalyst system, the effects of reaction temperature, EG/PET weight ratio, catalyst dosage, and reaction time on the glycolysis of PET were investigated in this paper. The effect of EG/PET weight ratio on the conversion of PET and the yield of BHET was shown in Fig. 6(a), according to the public work, the glycolysis reaction of PET occurred on the surface [38]. 7.5 g of EG was added to the reaction, the conversion rate of PET reached 90%, only 57.1% yield of BHET was obtained, which was mainly due to the influence of the concentration of EG. It is difficult to reach equilibrium between oligomer and BHET when the concentration of EG in the reaction was low, the increase in EG concentration was accompanied by an increase in BHET production. When the EG addition was 20 g, the concentration of DBU in the catalyst was affected by the high EG/PET contents, which reduced the conversion of the oligomer and affected the equilibrium conversion of the oligomer and BHET, and the conversion of PET and yield of BHET was decreased [25]. Therefore, the most suitable weight ratio of EG/PET was 3:1, which made the conversion of PET and yield of BHET reached maximum values, 96.93% and 78.2%, respectively.

The dosages of catalyst are significant factors in determining the reaction rate compared to the amount of solvent. As illustrated in Fig. 6 (b), when the molar ratio of catalyst dosage was increased from 1:1 to 1:6, the active sites on the PET macromolecular chain increased, thereby accelerating the breaking speed of PET [33]. The conversion and yield increased slowly and then reached a peak, in which the yield of BHET

was about 79.4% with 0.1 g DBU. Considering the effect of excess catalyst on the yield and economic cost of BHET, the optimal molar ratio Zn(OAc)₂/DBU was 1:2.

It could be concluded that temperature plays a crucial role in the PET glycolysis reaction and greatly influences the reaction rate of PET chain breakage. As depicted in Fig. 6(c), the yield of BHET increased from 60% to 78.2% when the temperature was 170 °C increased to 180 °C. The increase in temperature accelerated the expansion of PET and made it easier for the catalyst to attack the ester bonds in the molecular chain, but the yield decreased slowly when the temperature was up to 190 °C. The phenomenon can be explained that DBU was more active at low temperatures with the conversion of PET and the yield of BHET little changed when the reaction temperature was increased to 185–190 °C [25]. Therefore, 180 °C can be considered the most suitable temperature for glycolysis of PET.

Based on this, the influence of reaction time was examined in Fig. 6 (d), within 45 min of the reaction, the PET chains started to swell and the yields of BHET and PET were gradually increased as the reaction time extended. Most of the PET flakes were transformed into oligomers or BHET throughout the 45–77 min period. However, when the reaction time was elongated to 90 min, the conversion of PET up to 100%, while the yield of BHET decreased slightly. Herein, 77 min was regarded as the ideal glycolysis duration, the yield of BHET was at its highest values with 78.2%, the PET conversion rate reached the maximum value of 96.93%.

Compared with other metal salt catalysts [39,40], ionic liquids [23, 41] and other oxide catalysts [42], as demonstrated in Table 2. Combined different metal salt catalysts with DBU for PET glycolysis under

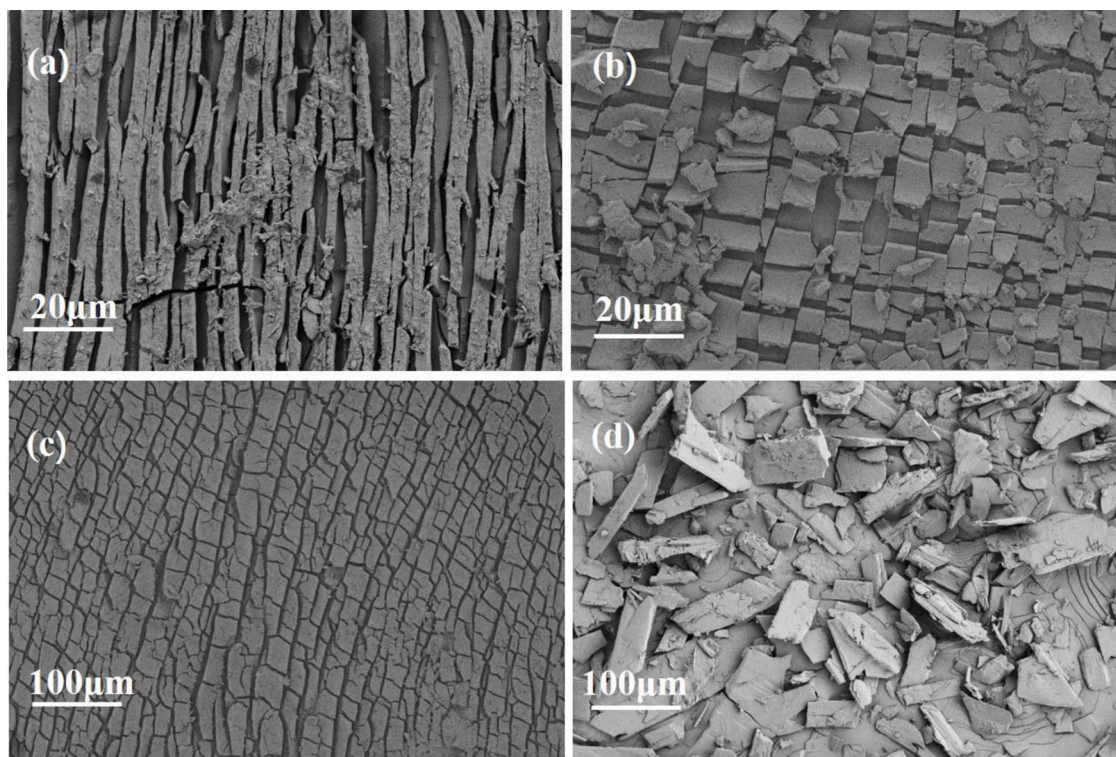


Fig. 5. SEM images of the residual PET surface after glycolysis time of (a), 20 min. (b), 40 min. (c), 60 min. (d), 77 min.

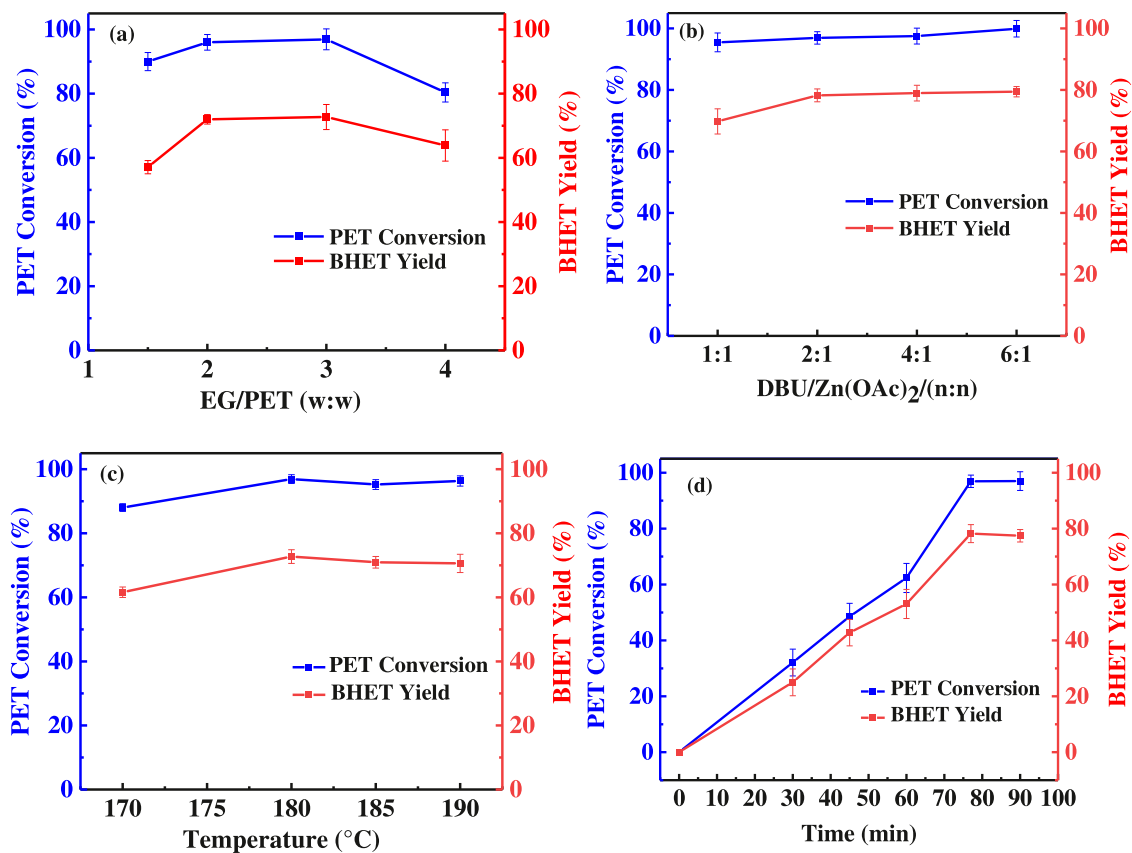


Fig. 6. (a), The effect of the weight ratio about EG/PET on the PET glycolysis reaction. Reaction conditions: $n(\text{Zn}(\text{OAc})_2)/\text{DBU} = 1:2$, $m(\text{Zn}(\text{OAc})_2) = 0.02$ g, 180°C for 77min. (b), The effect of the addition amount of the composite catalyst $\text{Zn}(\text{OAc})_2/\text{DBU}$ on the glycolysis reaction. Reaction conditions: PET 5 g, EG 15 g, 180°C for 77 min. (c), The effect of temperature on the PET glycolysis reaction. Reaction conditions: PET 5 g, EG 15 g, $n(\text{Zn}(\text{OAc})_2)/\text{DBU} = 1:2$, $m(\text{Zn}(\text{OAc})_2) = 0.02$ g, 77 min. (d), The effect of reaction time on the PET glycolysis reaction. Reaction conditions: PET 5 g, EG 15 g, $n(\text{Zn}(\text{OAc})_2)/\text{DBU} = 1:2$, $m(\text{Zn}(\text{OAc})_2) = 0.02$ g, 180°C .

Table 2

Comparison of glycolysis reaction conditions and catalytic activities of PET degradation.

Source	Catalysts	Temperature/(°C)	Time/(min)	PET conversion /(%)	BHETyield/(%)	Catalyst/PET weight ratios
This work	Zn(OAc) ₂ /DBU	180	77	100	78.2	0.004:0.006:1
	DBU	180	120	60.13	60	0.006:1
	Zn(OAc) ₂	180	120	44.83	45	0.004:1
Ref [39]	MnAc ₂	190	90	99.96	/	0.07:1
Ref [40]	ZnCl ₂	197	480	100	74	0.005:1
Ref [29]	Na ₂ CO ₃	196	60	/	78.8	0.006:1
	NaHCO ₃	196	60	/	60.2	0.003:1
Ref [33]	TBD	190	210	100	78	0.07:1
Ref [41]	[Bmim]Cl	190	120	28.9	6.0	0.05:1
	[Bmim]OH	190	120	100	71.2	0.05:1
	[Bmim]HCO ₃	190	120	82.8	14.1	0.05:1
Ref [20]	[Ch] [OAc]	180	240	98.2	85.2	0.05:1
	[Ch] [For]	180	180	94.3	82.9	0.05:1
	[Bmim]ZnCl ₃	190	120	100	82.6	0.05:1
Ref [19]	[Bmim]MnCl ₃	190	120	100	83.3	0.05:1
Ref [23]	γ-Fe ₂ O ₃	255	80	/	> 80	0.01:1
Ref [24]	GO-Mn ₃ O ₄	300	80	/	96.4	0.01:1
Ref [43]	Zn-MNPs	197	120	100	79.82	0.008:1

the same conditions was observed, and the best glycolysis effect was found for zinc acetate composite catalyst. The advantages of zinc ions over other metal salt ions were equivalent to Lewis acids coordinated to electron-rich oxygen atoms in the carbonyl group, making them more susceptible to nucleophilic attack. In the glycolysis reaction conditions, the combination of zinc acetate and DBU increased the reaction rate and reduced the content of metal salts for other different types of catalysts. The reaction time of this condition was shorter and glycolysis conditions were mild and simple, even the product has a higher purity.

3.6. Reaction kinetic of PET glycolysis

The phase change of PET macromolecules is a complex process, during the process of heating, catalysts and solvents occurred on the surface of the PET and outside-in PET will be broken down into oligomers or monomers. Influenced by many factors in the process of glycolysis, such as the shape and size of PET, the stirring rate and the way of glycolysis, and so on. It's reported that researchers have developed a variety of glycolysis dynamics models, such as the shrinking-core model [44,45], random scission model [46] and other mathematic models [47]. The SEM spectra (Fig. 5) showed that there are obvious fractures on the surface of the PET pellets, indicating that the glycolysis process of PET in EG was carried out on the surface of the PET raw material. Therefore, the kinetic reaction of glycolysis of PET was considered to be a first-order reaction, the kinetics model was assumed as the shrinking-core model initially and the kinetic reaction equation is as follows [48]:

$$\text{PET}_n + (n-1)\text{EG} \rightleftharpoons n\text{BHET} \quad (3)$$

$$\frac{d[\text{PET}]}{dt} = -k[\text{EG}][\text{PET}] \quad (4)$$

Due to the weight of EG in the reaction was much higher than that of PET. Therefore, the concentration of EG is regarded as a constant, k is the reaction rate constant, and $k[\text{EG}]$ is written as k' , the concentration of PET refers to the concentration of PET repeat units and x refers to the conversion rate of PET.

$$[\text{PET}] = [\text{PET}]_0(1-x) \quad (5)$$

$$k[\text{EG}] = k' \quad (6)$$

$$\frac{dX}{dt} = k'(1-x) \quad (7)$$

The above formula can be transformed into

$$\left[1 - (1-X)^{\frac{1}{n}}\right] = k''t \quad (8)$$

The glycolysis kinetics has mainly studied the effect of the reaction rate at different temperatures, as shown in Fig. 7(a), the reaction rate constants of the reaction at 170 °C, 180 °C, 185 °C and 190 °C were investigated, which were 0.05348 min⁻¹, 0.02657 min⁻¹, 0.02069 min⁻¹, 0.01024 min⁻¹, respectively. According to the Arrhenius equation (Eq. (9)), which was obtained in Fig. 7(b), where k'' is the reaction rate constant, E_a is the activation energy of the reaction, R is the gas constant, $R=8.314 \text{ J/(mol}\cdot\text{K)}$, and A is the pre-exponential factor;

$$\ln k'' = -\frac{E_a}{RT} + \ln A \quad (9)$$

The curve equation obtained by fitting was $y = -13.89x + 26.86$, and the apparent activation energy was calculated to be 115.48 kJ/mol. The result was consistent with the shrinking-core model assumed before. The activation energy of this energy was lower than the energy obtained by glycolysis of PET using zinc acetate (125 kJ/mol), the activation energy of ionic liquid degradation of PET was 162.82 kJ/mol [49], and the reaction activation energy of the PET glycolysis reaction mediated by HTBD-OAc/Zn(OAc)₂ DES was 131.31 kJ/mol [50], which demonstrated the effectiveness and significance of this composite catalyst for PET degradation [20].

3.7. Speculation on the reaction mechanism

It's known that the hydrogen bonds formed between EG and the catalysts have high catalytic activity in the glycolysis of PET [20,51]. The metal salts such as zinc acetate has been used as catalyst for glycolysis, and it is generally believed that the metal zinc ions attack the carbon groups on the ester bonds of PET, which leads to nucleophilic reactions and accelerates the breaking of ester bonds [8,52]. The process that DBU promotes the glycolysis of PET was that DBU should be seen as hydrogen bond acceptors since it is bases that can active ethylene glycol [53,54]. With comprehensive reference to their roles of in the glycolysis process, a possible collaborative catalyst degradation mechanism was proposed in this paper, where DBU formed hydrogen bond with the hydroxyl groups on ethylene glycol, and zinc is Lewis acid that can interact with carbonyl groups [32,55]. A probably reaction mechanism diagram is shown in Fig. 8, the molar ratio of Zn(OAc)₂/DBU was 1:2, which provided more attack points for the glycolysis reaction and accelerated the ester bond of PET. The synergistic catalysis of acids and bases in deep solvents (DESs) formed with metal salts in common ionic liquids have different mechanism [56].

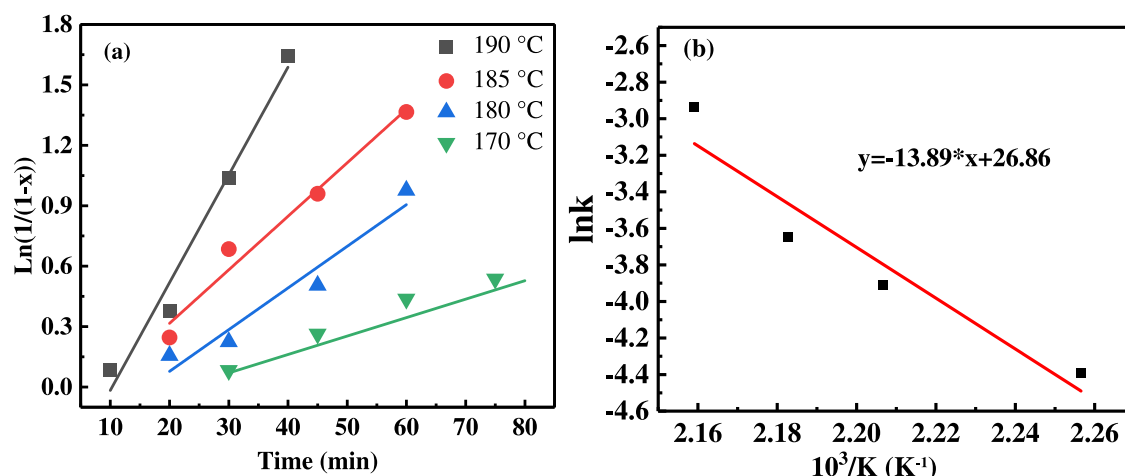


Fig. 7. (a), Effect of the temperature on the rate of PET glycolysis. (b), Arrhenius plot of the rate constant of PET glycolysis.

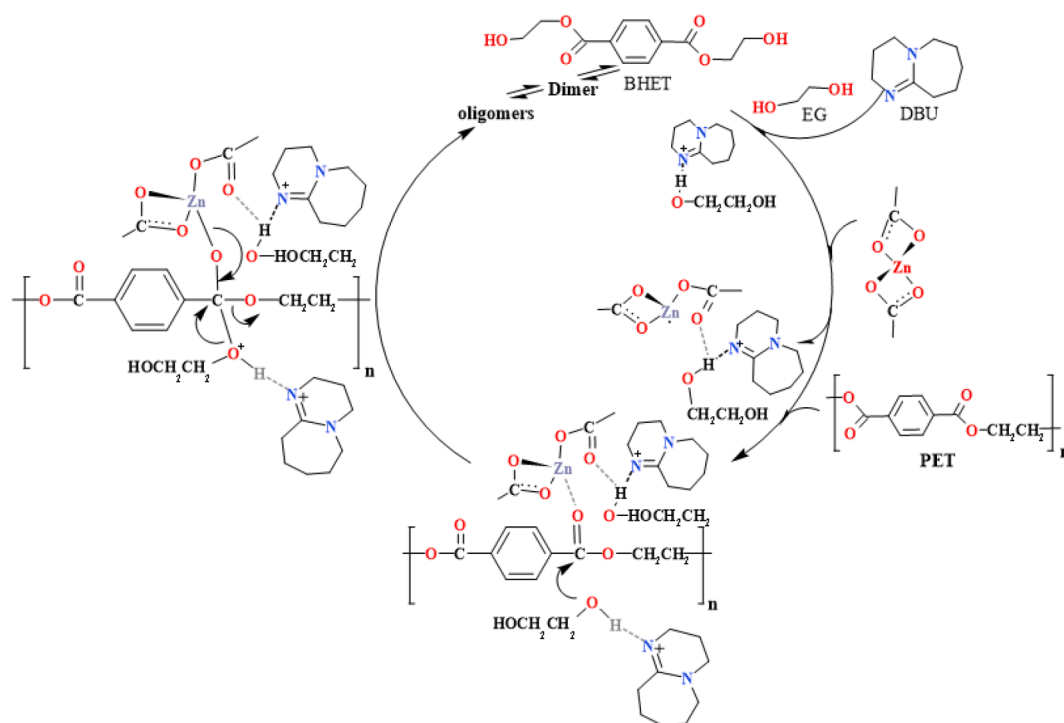


Fig. 8. Proposed mechanism for PET decomposition by combination of organocatalyst and metal salt.

4. Conclusion

This paper has explored the first composite catalysts of Zn(OAc)₂/DBU, after a series of reaction parameters analyzed, the optimal reaction conditions were obtained (180 °C, 77 min, m(PET: EG) = 1:3, n(Zn(OAc)₂/DBU = 1:2), m(Zn(OAc)₂: PET) = 0.004:1). Approaching 80% of BHET was achieved and the purity of the glycolysis product was 94.85%. The experiment indicated that the activation energy of the reaction was calculated to be 115.48 kJ/mol, which demonstrated the result was consistent with a shrinking-core model. It was speculated that the degradation mechanism was that DBU first forms a hydrogen bond donor with the hydroxyl group on ethylene glycol and accelerates the breakage of the PET molecular chain in combination with zinc acetate. The result of the work shows that recycling waste plastic without costly and sophisticated catalyst preparation, which have a positive impact on industrial perspectives and provide theoretical guidance.

CRediT authorship contribution statement

Wenyan Chen: Conceptualization, Investigation, Data curation, Writing – original draft, Formal analysis. **Maixin Li:** Data curation, Investigation, Formal analysis. **Xinchen Gu:** Investigation, Data curation. **Liang Jin:** Formal analysis, Resources. **Wenxing Chen:** Supervision, Project administration. **Shichang Chen:** Validation, Supervision, Project administration, Resources, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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